Modern Algorithmic Game Theory

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Recap

- Last week, we showed that Nash equilibria of two-player zero-sum games can be found using linear programming, and we relaxed Nash equilibria to correlated equilibria, which can be found using linear programming in general-sum games as well
- Linear programs give us exact solutions, but they need the whole payoff matrix, which makes them hard to scale to large games
- Fictitious Play is iterative, but it converges only in certain classes of games
- Today, we will introduce *regret*, a notion from online learning that provides worst-case guarantees for decisions made in an adversarial environment
- We will then use regret minimization as another self-play style algorithm, which converges to Nash equilibria in zero-sum games and to coarse correlated equilibria in general-sum games


Regret Minimization

Introduction to the Concept of Regret

We consider a problem of repeatedly making decisions in an uncertain environment.
We assume the following:

- A set of N actions $\mathcal{A}$, e.g. $\{\texttt{rock}, \texttt{paper}, \texttt{scissors}\}$
- An algorithm $\mathcal{H}$ that produces a probabilistic policy π at each timestep
- An adversarial environment that selects a loss vector l in response to the policy π

Examples:

- Choosing a road to take to work; the environment responds with traffic on roads
- Choosing an action in a game; the environment responds with an action

Introduction to the Concept of Regret

- We would like to study worst-case performance guarantees of our algorithm $\mathcal{H}$, even if the loss function is not known in advance and can be chosen arbitrarily by the environment
- We will use the notion of **regret** which provides strong guarantees in such settings
- Regret tells us how much we could have improved in **retrospect** if we had used an alternative policy π'; it is defined as the difference between the loss of our policy π and the alternative policy π'
- The alternative policy comes from a **comparison class** $\mathcal{G}$
- A comparison class consisting of individual actions $\mathcal{A}$ leads to the notion of **external regret**; we compare our performance to the best single action in retrospect
- There are also **internal** and **swap** regrets which allow richer comparison classes

Model Formalization

- We have a set of actions $\mathcal{A} = \{1, \ldots, N\}$
- At each time step t, the algorithm $\mathcal{H}$ chooses a policy $\pi^t \in \Delta(\mathcal{A})$
- The algorithm then receives a loss vector $l^t \in [0,1]^N$
- The algorithm's loss at time t is the weighted sum $l^t_{\mathcal{H}} = \sum_{a \in \mathcal{A}} \pi^t_a l^t_a$
- The cumulative loss of action a is $L^T_a = \sum_{t=1}^T l^t_a$, and the cumulative loss of the best action is $L^T_{min} = \min_{a \in \mathcal{A}} L^T_a$
- The algorithm's cumulative loss after T timesteps is simply $L^T = \sum_{t=1}^T l^t_{\mathcal{H}}$
- The external regret R^T_a of action a compares the cumulative loss that **would have been received** if we had played action a at each timestep t to the cumulative loss we have received. Mathematically, $R^T_a = L^T - L^T_a$
- Finally, the external regret of π is $R_\pi = \max_{a \in \mathcal{A}} R^T_a$

Regret with Respect to the Optimal Sequence

Theorem

Let $\mathcal{G}_{all}$ be a comparison class consisting of all functions mapping times $\{1, \ldots, T\}$ to actions $\mathcal{A} = \{1, \ldots, N\}$. Then, for any online algorithm $\mathcal{H}$, there exists a sequence of losses $l^1 \ldots l^T$, $l^t \in [0,1]^N$, such that the regret $R_{\mathcal{G}_{all}}$ is at least $T(1 - \frac{1}{N})$.

Proof: At each timestep t, the action a with the lowest probability π_a^t gets loss 0 and all the remaining actions get a loss of 1. Since $\min_{a \in \mathcal{A}} \pi_a^t \leq \frac{1}{N}$, the cumulative loss L^T of $\mathcal{H}$ will be at least $T(1 - \frac{1}{N})$. On the other hand, there exists an algorithm $g \in \mathcal{G}_{all}$ that achieves a total loss of 0.

- It is not possible to guarantee low regret with respect to the overall optimal sequence of decisions!

Restricted Comparison Class

- Now that we have shown we cannot hope to achieve a low overall regret against the optimal sequence of actions, we will restrict the comparison class
- We will look at a comparison class consisting of single actions, which is the external regret we defined earlier – we compare our performance to the best single action in hindsight
- We will first see that even this is out of reach for deterministic algorithms, and then show that stochastic algorithms can achieve it

Deterministic Algorithm $\mathcal{H}$

- Consider a deterministic algorithm $\mathcal{H}$ that places all of its probability mass on a single action a^t at each timestep t
- Now consider an algorithm that at each timestep t chooses an action a^t with the lowest cumulative loss $a^t = \arg\min_{a \in \mathcal{A}} L_a^{t-1}$
- The algorithm is also known as **Follow the Leader** (FTL) in the online optimization literature
- If each player in a repeated game uses FTL, we get the **Fictitious Play** algorithm
- Unfortunately, we can't guarantee a low total regret of any deterministic algorithm

Bound on the Loss of a Deterministic Algorithm

Theorem

For any deterministic algorithm $\mathcal{H}$, there exists a sequence of loss vectors for which $L^T = T$ while the best possible action in hindsight achieves $L^T_{min} \leq \frac{T}{N}$.

- Thus the worst-case total regret of any deterministic algorithm grows linearly, as it is at least $T(1 - \frac{1}{N})$, i.e. $\Omega(T)$
- Our goal is to find an algorithm that we call a **regret minimizer** that guarantees a **sub-linear** growth of the total regret, e.g. $\mathcal{O}(\sqrt{T})$
- This leads to the average regret approaching 0 as the number of timesteps $T \to \infty$
- Algorithms with guaranteed sub-linear total regret are sometimes called **no-regret** or **Hannan consistent** algorithms
- Can stochastic algorithms achieve sub-linear total regret?

Lower Bounds for Arbitrary Stochastic Algorithms

Theorem

Consider $T < \log_2 N$. There exists a stochastic generation of losses such that, for any online algorithm $\mathcal{H}$, we have $\mathbb{E}[L^T] = T/2$ and $L^T_{min} = 0$.

- In other words, the theorem says that we cannot hope to achieve sub-linear regret when the number of timesteps T is small compared to $\log_2 N$

Theorem

Let $N = 2$. There exists a stochastic generation of losses such that for any online algorithm $\mathcal{H}$ we have $\mathbb{E}[L^T - L^T_{min}] = \Omega(\sqrt{T})$.

- The theorem essentially says that even in the simplest scenario when there are only two actions, we cannot hope for regret $o(\sqrt{T})$

Regret Matching

- We would like to have an algorithm whose regret is close to these lower bounds
- There are many stochastic algorithms that achieve sub-linear regret, e.g. Polynomial Weights or Hedge
- One particularly simple and non-parametric algorithm is **Regret Matching**
- The algorithm considers only actions with positive regrets; therefore, we define $R_a^{t,+} = \max(R_a^t, 0)$
- At time t, the algorithm chooses all actions with non-zero $R_a^{t-1,+}$ with probability proportional to their value $R_a^{t-1,+}$
- $\pi_a^t = \frac{R_a^{t-1,+}}{\sum_{a' \in \mathcal{A}} R_{a'}^{t-1,+}}$ if $\sum_{a' \in \mathcal{A}} R_{a'}^{t-1,+} > 0$ and $\frac{1}{N}$ otherwise
- Regret Matching is guaranteed to have regret bounded by $\mathcal{O}(\sqrt{NT})$[1]

[1] See `http://www.cs.cmu.edu/~ggordon/ggordon.CMU-CALD-05-112.no-regret.pdf` for the proof.



Regret Minimization in Game Theory

Regret Minimization in Normal-Form Games

- Consider repeatedly playing a two-player normal-form game
- Instead of loss vectors, we will now switch to reward vectors
- Each player can employ a regret minimizer and consider its opponent as an adversarial environment
- The reward vector r_i^t can be computed using the opponent's policy π_{-i}^t and player i selects its next strategy π_i^{t+1} using its regret minimizer
- We define the average regret of player i after T timesteps as

$$\bar{R}_i^T = \frac{1}{T} \max_{a_i \in \mathcal{A}_i} \sum_{t=1}^{T} (u_i(a_i, \pi_{-i}^t) - u_i(\pi^t))$$

Convergence in General-Sum Games

- In general-sum games, if all players use a no (external) regret algorithm, the empirical distribution of actions converges to a **coarse correlated equilibrium**
- Coarse correlated equilibria are a generalization of correlated equilibria
- A coarse correlated equilibrium is a probability distribution over strategy profiles $a \in \mathcal{A} = \prod_{i \in \mathcal{N}} \mathcal{A}_i$, such that for all players $i \in \mathcal{N}$ and all unilateral deviations $a'_i \in \mathcal{A}_i$, it holds

$$\sum_{a \in \mathcal{A}} p(a) u_i(a) \geq \sum_{a \in \mathcal{A}} p(a) u_i(a'_i, a_{-i})$$

- If all players use a no-internal-regret algorithm, the empirical distribution of actions converges to a **correlated equilibrium**

Convergence in Zero-Sum Games

Theorem

Consider T iterations of a no-regret algorithm in a zero-sum game. If each player's average regret is less than ϵ then the **average** strategy profile $\bar{\pi}$ is a 2ϵ-Nash equilibrium.

Proof of Convergence in Zero-Sum Games

- Let π_1' be an arbitrary strategy of Player 1 and let $\bar{u} = \frac{1}{T}\sum_t u_1(\pi^t)$ be the average payoff of Player 1. Since the average regret of Player 1 is lower than ϵ and no strategy is better than the best action, we have

$$\frac{1}{T}\sum_t u_1(\pi_1', \pi_2^t) \leq \bar{u} + \epsilon$$

- The utility is linear in the strategy of each player, so taking the expectation over t gives us

$$u_1(\pi_1', \bar{\pi}_2) \leq \bar{u} + \epsilon \tag{1}$$

- Note that $\bar{u} \neq u_1(\bar{\pi})$ in general, as the utility is not linear in the whole strategy profile

Proof of Convergence in Zero-Sum Games

- Similarly, for an arbitrary strategy π_2' of Player 2, we have

$$u_2(\bar{\pi}_1, \pi_2') \leq \frac{1}{T} \sum_t u_2(\pi^t) + \epsilon$$

- Now, we will use our assumption that the game is zero-sum. We have $u_2 = -u_1$, which leads to

$$u_1(\bar{\pi}_1, \pi_2') \geq \bar{u} - \epsilon \tag{2}$$

- Chaining the two inequalities in Equations 1 and 2, we get

$$u_1(\pi_1', \bar{\pi}_2) - \epsilon \leq \bar{u} \leq u_1(\bar{\pi}_1, \pi_2') + \epsilon$$

Proof of Convergence in Zero-Sum Games

- This holds for all π_1' and π_2', so we can take the maximum and the minimum, respectively, which gives us

$$NashConv(\bar{\pi}) = \max_{\pi_1'} u_1(\pi_1', \bar{\pi}_2) - \min_{\pi_2'} u_1(\bar{\pi}_1, \pi_2') \leq 2\epsilon$$

- Finally, $\delta_i(\bar{\pi}) \geq 0$ for both players, so $\max_{i \in \mathcal{N}} \delta_i(\bar{\pi}) \leq NashConv(\bar{\pi}) \leq 2\epsilon$, i.e., $\bar{\pi}$ is a 2ϵ-Nash equilibrium

Solving Games with Regret Minimization

- We now have another iterative algorithm for solving normal-form games
- We can choose an arbitrary regret minimization algorithm, such as regret matching, and let both players play according to the algorithm
- If we choose regret matching, the asymptotic average regret for each player after T iterations is $\mathcal{O}(\frac{1}{\sqrt{T}})$
- When we then take the average strategy for each player, we have an $\mathcal{O}(\frac{1}{\sqrt{T}})$-Nash equilibrium
- For a fixed ϵ, we need $\mathcal{O}(\frac{1}{\epsilon^2})$ iterations of regret minimization

Properties of the Algorithm

- Very easy to implement
- Each player only needs to remember their cumulative regrets and their average strategy
- Players do not even have to know the payoff matrix – it does not have to be represented in the memory, as long as they observe the reward of every action in each round
- If the opponent does not play according to a regret minimization algorithm, we are guaranteed to earn as much as the best response to the opponent's average strategy in the limit

Convergence Notes

- We mentioned that the convergence rate of average regret of $\mathcal{O}(\frac{1}{\sqrt{T}})$ is optimal in the general setting
- However, in zero-sum games both players can cooperate to solve the game; therefore, they can achieve smaller regret and faster convergence
- There is a known algorithm with $\mathcal{O}(\frac{\ln T}{T})$ average regret for both players, with the assumption that they don't know the payoff matrix at the beginning[2]
- Optimistic Mirror Descent, which is closely related to the Exponential Weights algorithm, achieves the same rate under the same assumption[3]

[2] https://doi.org/10.1137/1.9781611973082.21
[3] https://arxiv.org/abs/1311.1869

Week 5 Homework

You can find more detailed descriptions of homework tasks in the GitHub repository.

1. Regret minimization